

Venkata Sai Revanth Damisetty

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Education

Cornell University, Ithaca, NY, United States.

Aug 2025 – May 2026

Masters in Applied Statistics (Data Science)

GPA : 3.66/4.0

Amrita Vishwa Vidyapeetham, Coimbatore, India.

Aug 2020 – May 2024

Bachelors in Computer & Communication Engineering (Department Rank: III)

GPA : 8.47/10

Work Experience

Applied Machine Learning Engineer (Founder-Led Project), PainMed PA, Austin, TX

Sep 2025 – Present

- **ERA Resubmission Automation:** Architected fault-tolerant ETL pipelines in Python/SQL with Pandas/NumPy over **X12-835** claims into PostgreSQL on AWS (EC2, S3, RDS), handling **10k+ claims/week**, and productionized RAG-based LLM inference services using LangGraph with ChromaDB embeddings and Dockerized CI/CD with validation guardrails to automate ERA resubmission workflows and shorten reimbursement turnaround time.
- **Market Signal Intelligence:** Engineered NLP ingestion and feature pipelines in Python/Spark on GCP (BigQuery), processing **~8k posts/month** to generate HuggingFace embeddings and apply unsupervised clustering and semantic similarity scoring, materializing topic and trend signals in a low-latency production analytics dashboard for sales outreach prioritization.

Machine Learning Engineer (Industry Client Collaboration), Los Alamos National Laboratory, NM

Jan 2026 – May 2026

- **LLM Classification Probability Evaluation Framework:** Built a reproducible **GenAI benchmarking pipeline** in Python/R to run prompt-variant sweeps across foundation models (GPT, Claude, Gemini, Perplexity), quantifying LLM performance and probability calibration with Brier/ECE and reliability curves under controlled data-generating processes within an automated experiment-tracking MLOps workflow.

Software Development Engineer (Edison Engineer), GE Healthcare, Bengaluru, India

Aug 2024 – Jun 2025

- **Hospital Compliance Automation:** Engineered and productionized a RAG-based semantic search service in Python/FastAPI using HuggingFace encoders and ChromaDB on AWS (EC2, S3), with Dockerized CI/CD deployment and latency-optimized retrieval and inference paths, reducing manual compliance questionnaire effort by **~40%**.

Software Development Engineer Intern, GE Healthcare, Bengaluru, India

Jan 2024 – Jun 2024

- **Dependency Risk Analysis Platform:** Architected dependency-graph ETL pipelines in Python/SQL over large code and binary artifacts with PostgreSQL for versioned metadata and MongoDB for dependency graphs, enabling scalable graph analytics to surface high-risk transitive mismatches and reduce post-release debugging time by **~30%**.

Publications

- [Optimizing ML Pipelines: Scalable Mobile-Based Tamil Sign Language Recognition](#), IEEE NKCon 2024
- [Smart Detection of Cyber Threats: A Machine Learning Approach](#), IJSREM 2024

Technical Skills

Programming & Frameworks: Python, SQL, PyTorch, TensorFlow, scikit-learn, LangChain, LangGraph

Data & Systems: NumPy, Pandas, PostgreSQL, MongoDB, ChromaDB, REST APIs, Docker, Linux, CI/CD

Cloud: AWS (EC2, S3, RDS), GCP (BigQuery)

Competencies: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Agentic Workflows (Agentic AI), Model Evaluation, Feature Engineering, Unsupervised Learning, Statistical Modeling, ETL Pipelines, MLOps (experiment automation)

Projects

SignStream — Interactive ASL Recognition System

- Architected and deployed a real-time deep learning inference service (Python, PyTorch, TensorFlow, REST APIs) with feature engineering on MediaPipe landmarks, containerized via Docker on Linux with CI/CD, achieving **92.3% accuracy** and **<150 ms** end-to-end latency through optimized model training, evaluation, and deployment pipelines.

Certifications

- [Machine Learning Specialization](#) – Stanford
- [Generative AI for Data Scientists](#) – IBM
- Certified ML Engineer Associate - AWS